

COLDPOD · REQUIREMENTS

ColdPod requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/coldpod		
Source	docs/03-requirements.md @ f07b29f		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

ColdPod requirements

These are first-pass requirements for the concept. Targets are proposals for review, awaiting Amish, and will be checked by calculation at TRL 3. The status column compares each target with the first-order estimates in CPD-PRC-001; three requirements are **not met** by the current concept (R6, R12 and R15) and two are met with thin margins (R4 and R5).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status (estimate)
R1	Keep the payload in the safe range	2 to 8 °C at a buffered probe anywhere in the payload space; control setpoint 5 °C	Heat-transfer calculation; later lab test in a climate chamber	Met by design
R2	Never freeze the payload	No payload-contact surface below 1 °C in normal operation or with a stuck-on Peltier driver	Calculation of the cold path; fault analysis	Met by design (5 °C PCM between cold path and liner; hardware cut-out)
R3	Carry a useful load	1.0 L or more of usable payload; holds 20 or more insulin pens up to 170 mm long	Massing model	Met: about 1.2 L, about 24 pens
R4	Hold with no power at all	12 h or more at a constant +43 °C, starting with the PCM frozen	Heat-leak and PCM calculation	Met, thin margin: about 12.7 h
R5	Hold off-grid on a full charge	24 h or more at a constant +32 °C on the internal battery, then the PCM	Energy budget	Met, thin margin: about 26 h
R6	Hold off-grid on a full charge in extreme heat	24 h or more at a constant +43 °C on the internal battery, then the PCM	Energy budget	Not met: about 16 h
R7	Hold indefinitely on external power	Keeps 2 to 8 °C up to +43 °C from 12 V DC (10 to 15 V) or USB-C PD (20 V, 45 W or more)	Peltier and heat sink calculation	Met on estimate (about 17 W needed at 43 °C)
R8	Recharge the cold quickly	Refreeze a fully melted PCM in 8 h or less at 25 °C ambient, box empty, while charging the battery	PCM and Peltier calculation	Met: about 6 h
R9	Record the temperature	Payload probe accuracy ± 0.5 °C; log every 1 min; 60 days or more on board; export as CSV over Bluetooth Low Energy or USB	Datasheet and storage calculation; later calibration	Met by design
R10	Raise alarms early	Local sound, light and display alarm, plus a phone notification when paired: warn after 10 min outside 2 to 8 °C; alarm at once at 0 °C or lower; low battery, sensor fault and lid open for more than 2 min	Design review of firmware sketch	Met by design
R11	Keep logging when the cooling stops	Logger runs 14 days or more after the Peltier is shut off for low battery	Power budget	Met: reserve lasts weeks
R12	Be light enough to carry all day	5.0 kg (11 lb) or less empty	Mass estimate, later weighing	Not met: about 5.2 kg
R13	Be compact	Fits in 400 x 250 x 250 mm including handle	Massing model	Met: about 370 x 215 x 205 mm
R14	Travel by air	Battery 100 Wh or less, removable or with a shipping switch	Battery specification	Met: 76.8 Wh

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status (estimate)
R15	Be affordable and buildable	Parts cost \$250 or less; no custom PCB for the first build	Priced BOM	Not met: about \$280
R16	Survive field use	Splash resistant (IP54 target for electronics bays); survives a 0.5 m drop onto a hard floor while loaded	Design review; later test	Unverified
R17	Be clearly labelled as a prototype	"Research prototype, not a medical device" on the box, the display start screen and every exported log	Design review	Met by design

■ Assumptions

- Ambient cases follow the WHO PQS test point of +43 °C for the hot zone ([WHO PQS E004/VC01](#)), +32 °C for a typical hot day and 25 °C for a clinic.
- The 2 to 8 °C window and the no-freezing rule come from insulin and vaccine storage guidance ([US FDA](#); [CDC Pink Book, chapter 5](#)).
- Logging and alarm targets are stricter than a WHO 30-day recorder (−0.5 °C for 60 min, +8 °C for 10 h; [TechNet-21](#)) because a carrier in transit can move out of range much faster than a refrigerator.
- R6 matches the WHO long-range carrier cold life only loosely: a WHO carrier is tested passive for 30 h at 43 °C. ColdPod does not claim WHO equivalence.
- The alarm thresholds in R10 are proposed defaults, awaiting Amish, and will be adjustable per product.